

Job Market Intelligence Project – End-to-End Summary

1. Project Overview

This project analyses job posting data to understand hiring behaviour, company dominance, industry demand, and remote work strategies. The goal is to transform raw job posting data into **actionable business insights** using **Excel, SQL, and analytical thinking**, and present them through an interactive dashboard suitable for interviews and real-world decision-making.

The project follows a realistic analytics workflow: - Raw data ingestion - Metric engineering - SQL-style logic translated into Excel - Pivot-based analysis - What-if and scenario analysis - Dashboard storytelling

2. Data Source

The core dataset is a job postings file (`postings.csv / postings.xlsx`) containing: - Company name - Job postings (individual rows) - Indicators for remote roles - Industry and role classifications (derived)

Each row represents a **single job posting**, not a company. All company-level insights are **aggregated**, mimicking real analytics work.

3. Data Preparation & Feature Engineering

3.1 Company-Level Metrics (Excel_Enriched Sheet)

Raw job postings are aggregated at the **company level** to create analytical features:

- **Total Jobs** – total number of postings per company
- **Remote Jobs** – number of postings marked as remote
- **Remote Share %** – $\text{Remote Jobs} \div \text{Total Jobs}$

This step converts raw operational data into decision-ready metrics.

3.2 Derived Business Labels

To move beyond simple counts, business logic is applied:

Hiring Category - Low Hiring - Medium Hiring - High Hiring

Remote Strategy - Onsite-Heavy - Hybrid - Remote-First

These labels simulate how analysts translate numbers into executive-friendly language.

4. Pivot Table Analysis

Pivot tables are the analytical backbone of the project. All dashboard elements are driven by pivots, ensuring scalability and slicer control.

4.1 Key Pivot Analyses

1. **Top Hiring Companies**

2. Identifies companies with the highest hiring volume

3. Reveals market dominance

4. **Top Industries**

5. Aggregates job volume by industry

6. Highlights sectors driving employment demand

7. **Hiring Strategy Mix**

8. Distribution of Onsite, Hybrid, and Remote-First companies

9. Provides workforce strategy insights

10. **Hiring Intensity Breakdown**

11. Shows how many companies are aggressively vs lightly hiring

12. **Remote Share vs Total Jobs**

13. Compares hiring scale with remote adoption

14. Identifies strategic outliers

5. What-If & Scenario Analysis

To demonstrate advanced analytical thinking, the project includes **what-if analysis**.

5.1 Sensitivity Analysis

Scenarios model how changes in remote hiring affect overall strategy: - Current state - Mild shift (+10% remote) - Moderate shift (+30% remote) - Aggressive shift (+60% remote)

Each scenario recalculates: - New Remote Jobs - New Remote Share % - Resulting Remote Strategy classification

This shows how small operational changes can alter strategic positioning.

5.2 Goal Seek

Goal Seek is used to answer questions like:

"How many remote jobs are needed to become a Hybrid or Remote-First company?"

This mirrors real business planning and policy evaluation.

6. Dashboard Design

The **Job Market Intelligence Dashboard** is the final output and storytelling layer.

6.1 Core KPIs

- **Total Jobs** – overall market activity
- **Top Hiring Company** – market leader
- **Top Industry** – highest demand sector

6.2 Interactive Elements

- Company slicer
- Industry slicer
- Hiring volume filters

All charts update dynamically, allowing exploratory analysis.

6.3 Visual Components

- Bar charts for hiring dominance
 - Donut and pie charts for strategy mix
 - Line and trend visuals for hiring patterns
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7. Business Insights

Key insights derived from the dashboard include:

- Hiring activity is **highly concentrated**, with a small number of companies dominating job volume.
 - **Tech Recruitment and IT Consulting** consistently lead industry demand.
 - Most companies follow an **onsite-heavy or hybrid model**, indicating cautious adoption of fully remote work.
 - High-volume hiring companies tend to rely less on remote roles, suggesting operational or compliance constraints.
 - Market growth is driven more by **many small and medium hiring firms** than by uniform large-scale expansion.
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8. Tools & Skills Demonstrated

- Excel (Advanced): Pivot Tables, Slicers, What-If Analysis, Goal Seek
 - SQL thinking translated into Excel logic
 - KPI design and metric engineering
 - Business storytelling and dashboard design
 - Scenario planning and sensitivity analysis
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9. Interview Value

This project demonstrates: - End-to-end analytical thinking - Ability to convert raw data into executive insights - Strong understanding of hiring, workforce strategy, and market dynamics - Practical, job-ready Excel and analytics skills

It is suitable for roles in: - Data Analysis - Business Intelligence - Operations Analytics - Market Research

10. Conclusion

This Job Market Intelligence project goes beyond basic reporting. It combines **data preparation, analytical logic, scenario modelling, and dashboard storytelling** to deliver insights that mirror real-world business analysis.

It is not just an Excel project—it is a complete analytics case study.